

# YUGESH BHADWAL

Automation Engineer | AI & Generative AI Specialist

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## PROFESSIONAL SUMMARY

Automation & AI Engineer with 3+ years of experience building enterprise automation solutions and machine learning-powered AI applications. At Mercedes-Benz Research and Development India (MBRDI), built and deployed 20+ automation workflows using UiPath, Power Automate, VBScript, and Python — integrating with SAP and enterprise applications to save 200+ hours per week. Independently designed and shipped production-grade AI systems: a multi-agent NLP-driven data analyst (LangChain, LangGraph, Azure OpenAI) and an enterprise RAG knowledge assistant with vector database-backed semantic search. Proficient across the full AI/ML development stack: large language models (LLMs), prompt engineering, embeddings, RAG pipelines, agentic workflows, REST API development, and MLOps-ready deployment practices (Docker, CI/CD). Holds Microsoft Azure AI Fundamentals certification.

## PROFESSIONAL EXPERIENCE

### Automation Engineer

*Mercedes-Benz Research and Development India (MBRDI) | Bengaluru, India | 2023 – Present*

#### Key Responsibilities

- Designed and implemented 20+ automation solutions to improve operational efficiency, saving 200+ hours per week in manual effort across business processes.
- Developed and maintained end-to-end workflow automations using **UiPath** (Robotic Process Automation), **Power Automate** & **Power Automate Desktop**, **VBScript**, and custom **Python** scripts to automate complex business processes.
- Built automations integrating with **SAP** (including SAP GUI scripting and SAP data extraction workflows), alongside other enterprise applications such as Microsoft 365 suite and internal business systems.
- Integrated enterprise applications through REST APIs and automated data exchange processes, enabling seamless cross-system workflows.
- Collaborated with business stakeholders to gather requirements, analyze processes, and deliver scalable automation solutions.
- Created monitoring, reporting, and exception-handling mechanisms to improve automation reliability and uptime.
- Worked on data extraction, transformation, validation, and reporting workflows involving structured and semi-structured data sources.

#### Key Achievements

- Delivered 20+ automation workflows end-to-end, collectively saving 200+ hours per week in manual processing effort.
- Led migration of 200+ UiPath bots to Power Automate, personally delivering 20 workflows; improved maintainability, reduced licensing costs, and ensured business continuity during transition.
- Automated critical SAP-integrated workflows (data extraction, report generation, order processing triggers) that previously required significant manual intervention.
- Built reusable automation components and frameworks in Python and Power Automate that accelerated onboarding of migrated workflows and reduced future development effort.
- Developed VBScript-based automation utilities to supplement RPA bots for legacy application interactions and Windows-level process automation.

## PERSONAL PROJECTS

### Agentic Conversational Data Analyst

*Technologies: Python, LangChain, LangGraph, Azure OpenAI, SQLite/PostgreSQL, FastAPI, Streamlit | GitHub: [github.com/yugesh58/sql-ai-platform](https://github.com/yugesh58/sql-ai-platform)*

- Developed an AI-powered NLP-driven conversational analytics platform that translates natural language into SQL queries, reducing ad-hoc data request turnaround from hours to seconds.

- Implemented semantic schema retrieval using vector embeddings and FAISS for intelligent context-aware querying; achieved ~90% query accuracy on benchmark test cases.
- Built multi-step agentic workflows using LangGraph for query planning, SQL generation, validation, execution, and result interpretation — exposing results via a REST API (FastAPI).
- Integrated Azure OpenAI Service (GPT-4) as the LLM backbone; added conversational memory and follow-up query handling for natural multi-turn interactions with structured data.

## Enterprise RAG Knowledge Assistant

*Technologies: Python, LangChain, Azure OpenAI, Hugging Face, Vector Databases, Docker, PDF Processing | GitHub: [github.com/yugesh58/enterprise-ai-platform](https://github.com/yugesh58/enterprise-ai-platform)*

- Built a production-ready Retrieval-Augmented Generation (RAG) system for querying enterprise documents; reduced time-to-answer for knowledge lookups by an estimated 70% vs manual search.
- Designed end-to-end NLP data pipelines: document ingestion, chunking, embedding generation (Hugging Face sentence-transformers), and vector search with relevance ranking.
- Enabled context-aware question answering across 1,000+ page document collections using semantic search, cutting irrelevant results by over 60% compared to keyword-only search.
- Containerised the full application with Docker for MLOps-friendly deployment; designed a scalable architecture suitable for enterprise knowledge management use cases.

## TECHNICAL SKILLS

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|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming:</b>                  | Python, SQL, VBScript, C++                                                                                                                                  |
| <b>AI &amp; Machine Learning:</b>    | Generative AI, Agentic AI, Machine Learning, NLP, Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-tuning, Vector Databases, Embeddings, LLMs |
| <b>Frameworks &amp; Libraries:</b>   | LangChain, LangGraph, Azure OpenAI, Hugging Face, FastAPI, Streamlit, Pandas, FAISS                                                                         |
| <b>Databases:</b>                    | PostgreSQL, SQLite                                                                                                                                          |
| <b>Automation &amp; Integration:</b> | Power Automate, Power Automate Desktop, Copilot Studio, UiPath, SAP GUI Scripting, REST API Development, Workflow Automation, Process Optimization          |
| <b>DevOps &amp; MLOps:</b>           | Docker, CI/CD Pipelines (GitHub Actions), MLOps                                                                                                             |
| <b>Cloud:</b>                        | Microsoft Azure (Azure OpenAI Service, Azure AI Fundamentals)                                                                                               |
| <b>Tools &amp; Platforms:</b>        | Git, GitHub, VS Code                                                                                                                                        |

## EDUCATION

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### Bachelor of Engineering (B.E.) – Information Science and Engineering

*NMAM Institute of Technology (NMAMIT) | Nitte, Karnataka, India | 2019 – 2023*

## CERTIFICATIONS & LEARNING

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- Microsoft Azure AI Fundamentals (AI-900)
- Microsoft GitHub Actions